The exact DLV–Black product lift from the Wiese–Wissel drift condition to a smooth SANOS surface

A self-contained construction and proof

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Abstract

This note proves the particular construction of interest. The factor X is the discrete martingale encoded by a dynamically arbitrage-free DLV density tower. Its DLV coordinates obey the Wiese–Wissel drift condition: Wiese's density-coordinate form and Wissel's price-coordinate form are algebraically equivalent. An independent positive Black–Scholes martingale Y supplies the smoothing. For $S = XY$, the conditional call surface is a finite SANOS mixture. An exact Itô calculation proves zero drift not only at the inherited DLV nodes but at every strike between them. The proof also identifies the necessary orthogonality, maturity-clock and grid-recalibration qualifications.

1 What is being identified

The roles of the two factors are different:

X :	the discrete martingale factor generated by the DLV density tower,
Y :	an independent positive Black martingale generating the smooth kernel,
$S = XY$:	the observable smoothed underlying.

(1)

Strictly speaking, X is not "the DLV drift". The DLV model determines the conditional laws of X and the dynamics of the DLV coordinates. The Wiese–Wissel condition determines the risk-neutral drift of the local variance coordinate ℓ. The stock factor X itself is a martingale.

We work in discounted units. Let

$$0 < x_0 < \cdots < x_N$$

be the latent DLV support and let $T_1 < \cdots < T_M$ be the model expiries. A coherent DLV tower has conditional masses

$$q_i^m(t) = \mathbb{Q}(X_{T_m} = x_i \mid \mathcal{F}_t^X), \qquad t \le T_m, \tag{2}$$

so that

$$q_i^m \ge 0, \qquad \sum_i q_i^m = 1, \qquad \sum_i x_i q_i^m = X_t. \tag{3}$$

For fixed (m, i), $q_i^m(t)$ is a bounded martingale. In the continuous case write

$$\mathrm{d}q_i^m(t) = \psi_i^m(t) \cdot \mathrm{d}W_t^X, \qquad \mathrm{d}X_t = \sum_i x_i \psi_i^m(t) \cdot \mathrm{d}W_t^X. \tag{4}$$

The second formula is independent of m by the common-forward condition.

The hypothesis that the DLV tower is coherent is material. It is supplied, for example, by a DLV/Wissel model satisfying its existence assumptions, or by finite martingale transition kernels between the expiry marginals. A list of unrelated static density fits is not yet such a process.

2 The X-only DLV drift condition

Define the X-only calls on a fixed cash strike grid K_n by

$$c_n^m(t) = \mathbb{E}[(X_{T_m} - K_n)^+ \mid \mathcal{F}_t^X] = \sum_i q_i^m(t)(x_i - K_n)^+. \tag{5}$$

Every c_n^m is a martingale. Its discrete butterfly is

$$\widehat{q}_n^m = \frac{c_{n+1}^m - c_n^m}{K_{n+1} - K_n} - \frac{c_n^m - c_{n-1}^m}{K_n - K_{n-1}}. \tag{6}$$

When the strike and support grids coincide, with the usual boundary masses, $\widehat{q}^m$ is precisely the conditional mass vector in (2). We retain a hat in this section to show which fact is used: $\widehat{q}_n^m$ is a deterministic linear combination of traded calls and therefore has zero drift.

For a back-month slab define ℓ_n^m by the exact discrete Dupire identity

$$c_n^m - c_n^{m-1} = \omega_n^m \ell_n^m \widehat{q}_n^m, \qquad \omega_n^m = \frac{K_n^2(T_m - T_{m-1})}{K_{n+1} - K_{n-1}}. \tag{7}$$

Assume $\widehat{q}_n^m > 0$ and write

$$\mathrm{d}\widehat{q}_n^m = \psi_n^m \cdot \mathrm{d}W_t^X, \qquad \mathrm{d}\ell_n^m = \alpha_n^m \,\mathrm{d}t + \beta_n^m \cdot \mathrm{d}W_t^X. \tag{8}$$

Proposition 2.1 (Wiese–Wissel drift condition)**.** *For every back-month node,*

$$\boxed{\alpha_n^m = -\frac{\beta_n^m \cdot \psi_n^m}{\widehat{q}_n^m} = -\eta_n^m \cdot \beta_n^m, \qquad \eta_n^m := \frac{\psi_n^m}{\widehat{q}_n^m}.} \tag{9}$$

Under the exact identification $\ell_n^m = (x_n^m)^2$, this is the price-coordinate form of the Wiese–Wissel risk-neutral drift condition.

Proof. The left-hand side of (7) is the difference of two martingales. The drift of its right-hand side is

$$\omega_n^m(widehatq_n^m\alpha_n^m + \beta_n^m \cdot \psi_n^m).$$

It must vanish. Division by $\omega_n^m \widehat{q}_n^m$ proves (9).

For the price-space dictionary, suppose Wissel writes

$$\frac{\mathrm{d}x_n^m}{x_n^m} = \bar{u}_n^m \,\mathrm{d}t + v_n^m \cdot \mathrm{d}W_t^X.$$

Since $\ell = (x)^2$, Itô's formula gives

$$\beta_n^m = 2\ell_n^m v_n^m, \qquad \frac{\alpha_n^m}{\ell_n^m} = 2\bar{u}_n^m + \|v_n^m\|^2.$$

The relative calendar-spread volatility is

$$R_n^m = \frac{\beta_n^m}{\ell_n^m} + \frac{\psi_n^m}{\widehat{q}_n^m} = 2v_n^m + \eta_n^m.$$

Substitution in Wissel's risk-neutral condition $\bar{u}_n^m = (-R_n^m + \frac{3}{2}v_n^m) \cdot v_n^m$ gives

$$\frac{\alpha_n^m}{\ell_n^m} = -2\eta_n^m \cdot v_n^m = -\eta_n^m \cdot \frac{\beta_n^m}{\ell_n^m},$$

which is exactly (9). □

Thus Wiese's density-coordinate derivation and Wissel's price-coordinate derivation impose the same grid drift. We call their common restriction the *Wiese–Wissel drift condition*. This is the drift restriction inherited by the X factor in the product construction.

With jumps, the exact predictable-drift condition is

$$\alpha_n^m = -\frac{\beta_n^m \cdot \psi_n^m + \int \Delta\ell_n^m(z)\Delta\widehat{q}_n^m(z)\,\nu_t(\,\mathrm{d}z)}{\widehat{q}_n^m}. \tag{10}$$

The continuous formula is sufficient for the smooth lift proved next.

3 The independent Black smoothing factor

Let W^Y be Brownian motion independent of $\mathcal{F}_\infty^X$, and set

$$\frac{\mathrm{d}Y_t}{Y_t} = \sigma_Y(t)\cdot\,\mathrm{d}W_t^Y, \qquad Y_0 = 1, \tag{11}$$

where σ_Y is deterministic and square-integrable. Put

$$V(t) = \int_0^t \|\sigma_Y(u)\|^2\,\mathrm{d}u, \qquad w(t,T) = V(T) - V(t). \tag{12}$$

Then Y is a positive true martingale and, conditionally on $\mathcal{F}_t^Y$,

$$\frac{Y_T}{Y_t} = \exp\!\left(-\frac{1}{2}w(t,T) + \sqrt{w(t,T)}Z\right), \qquad Z\sim N(0,1).$$

Let

$$B(a,K,w) = \mathbb{E}[(ae^{-w/2+\sqrt{w}Z} - K)^+] \tag{13}$$

denote the Black call kernel. It satisfies the heat identity

$$B_w(a,K,w) = \frac{1}{2}a^2 B_{aa}(a,K,w). \tag{14}$$

The function B is the smoothing operator; Y is the martingale whose conditional transition law produces that operator.

4 Exact product-lift theorem

Theorem 4.1 (DLV–Black product lift)**.** *Let X be a positive, bounded DLV martingale with coherent conditional density tower* (2)*, whose back-month coordinates satisfy* (9)*. Let Y be the independent Black martingale* (11)*, and define*

$$S_t = X_t Y_t. \tag{15}$$

Then S is a positive true martingale. For every model expiry T_m, every trading time $t \le T_m$, and every cash strike $K > 0$, its call price is

$$\boxed{C_t(T_m,K) = \sum_{i=0}^{N} q_i^m(t)B(x_i Y_t, K, w(t,T_m))\,.} \tag{16}$$

Consequently:

(i) (16) *is a finite SANOS representation with weights* q_i^m*, anchors* $A_i(t) = x_i Y_t$*, and common smoothing variance* $w(t, T_m)$*;*

(ii) it is statically arbitrage-free in strike and maturity;

(iii) for every fixed (T_m, K)*, including every* K *strictly between DLV nodes,* $C_t(T_m, K)$ *is a true* $\mathbb{Q}$*-martingale;*

(iv) at expiry it settles to the ordinary payoff $(S_{T_m} - K)^+$*.*

Proof. Independence and positivity give

$$\mathbb{E}[S_T \mid \mathcal{F}_t^X \vee \mathcal{F}_t^Y] = \mathbb{E}[X_T \mid \mathcal{F}_t^X]\mathbb{E}[Y_T \mid \mathcal{F}_t^Y] = X_t Y_t = S_t,$$

so S is a true martingale. Condition first on $X_{T_m} = x_i$ and then on $\mathcal{F}_t^X \vee \mathcal{F}_t^Y$. The conditional law of Y_{T_m}/Y_t is lognormal with total variance $w(t, T_m)$, hence

$$\mathbb{E}[(S_{T_m} - K)^+ \mid \mathcal{F}_t] = \sum_i q_i^m(t) B(x_i Y_t, K, w(t, T_m)),$$

which proves (16). It is a conditional expectation of an ordinary call payoff on a single martingale. Dynamic no-arbitrage and calendar monotonicity follow. Positivity of the weights and the Black kernels give monotonicity and convexity in strike.

At $t = T_m$, $w(T_m, T_m) = 0$, and the conditional mass collapses onto the realised state $X_{T_m} = x_I$. Therefore

$$C_{T_m}(T_m, K) = B(x_I Y_{T_m}, K, 0) = (S_{T_m} - K)^+.$$

□

The theorem gives the exact construction requested: the DLV factor supplies the state-dependent mixture weights and their constrained dynamics; the Black factor supplies the common smoothing; their product supplies one underlying whose entire smooth call surface is dynamically arbitrage-free.

5 The between-grid drift condition, calculated explicitly

The conditional-expectation proof is decisive, but the generator calculation shows exactly how the grid drift is extended. Fix (T_m, K) and abbreviate

$$a_i(t) = x_i Y_t, \qquad w_m(t) = w(t, T_m), \qquad G_i(t) = B(a_i(t), K, w_m(t)).$$

Since

$$\mathrm{d}a_i = a_i \sigma_Y \cdot \mathrm{d}W^Y, \qquad \mathrm{d}w_m = -\|\sigma_Y\|^2 \,\mathrm{d}t,$$

Itô's formula and (14) give

$$\begin{aligned} \mathrm{d}G_i &= B_a(a_i, K, w_m) a_i \sigma_Y \cdot \mathrm{d}W^Y \\ &\quad + \left[-\|\sigma_Y\|^2 B_w + \frac{1}{2}\|\sigma_Y\|^2 a_i^2 B_{aa} \right] \mathrm{d}t \\ &= B_a(a_i, K, w_m) a_i \sigma_Y \cdot \mathrm{d}W^Y. \end{aligned} \tag{17}$$

Thus every moving Black kernel is itself a martingale. Now

$$C_t(T_m, K) = \sum_i q_i^m(t) G_i(t). \tag{18}$$

Integration by parts gives

$$\mathrm{d}C = sum_i G_i \ \mathrm{d}q_i^m + \sum_i q_i^m \ \mathrm{d}G_i + \sum_i \mathrm{d}\langle q_i^m, G_i\rangle_t. \tag{19}$$

The first two sums have zero drift. Independence gives $\mathrm{d}\langle W^X, W^Y\rangle_t = 0$, so the last sum vanishes. Hence

$$\boxed{\text{drift } C_t(T_m, K) = 0 \quad \text{for every } K > 0.} \tag{20}$$

Equivalently, the full off-grid Itô restriction is

$$\boxed{\sum_i B_i \, \mu_{q_i} + \sum_i q_i \left(-\|\sigma_Y\|^2 B_{w,i} + \tfrac{1}{2}\|\sigma_Y\|^2 a_i^2 B_{aa,i}\right) + \sum_i a_i B_{a,i}, \frac{\mathrm{d}\langle q_i, \log Y\rangle_t}{\mathrm{d}t} = 0.} \tag{21}$$

In the exact construction, its three lines vanish respectively because

$$\mu_{q_i} = 0, \qquad B_w = \tfrac{1}{2} a^2 B_{aa}, \qquad \mathrm{d}\langle q_i, \log Y\rangle = 0. \tag{22}$$

Equation (21) is the desired between-node extension of the finite DLV drift condition. It is valid for every strike, not only for a dense numerical audit mesh.

If W^X and W^Y are correlated, the last term in (21) is generally nonzero. The simple product lift then requires an additional compensating drift. Independence, or at least orthogonality of the relevant martingale parts, is therefore a structural condition rather than a cosmetic assumption.

6 How the node condition and the smooth condition fit together

There are two exact but distinct statements.

First, on the latent X grid, the DLV identity

$$c_n^m - c_n^{m-1} = \omega_n^m \ell_n^m \widehat{q}_n^m$$

and its product rule give the Wiese–Wissel drift condition

$$\alpha_n^m = -\eta_n^m \cdot \beta_n^m.$$

Second, after multiplication by Y, the actual smoothed calls satisfy

$$C_t(T_m, K) = \sum_i q_i^m B(x_i Y_t, K, w_m(t))$$

and the extended drift identity (21) vanishes for every strike. If one samples these actual smooth calls on any strike grid and recomputes their butterflies and DLV coordinates, those newly computed coordinates also satisfy the Wiese–Wissel drift condition, because they are deterministic functions of martingale calls. They need not be numerically equal to the latent X-only DLV coordinates.

This last point is essential. Multiplying an already calibrated X by a nontrivial Y generally changes the prices at the original cash strikes:

$$\sum_i q_i^m B(x_i Y_t, K_n, w_m(t)) \neq \sum_i q_i^m (x_i - K_n)^+. \tag{23}$$

Therefore the product theorem is an exact dynamic construction, but it is not an automatic price-preserving post-processing of an arbitrary pre-existing DLV grid.

To match a prescribed finite call grid $C_n^{m,\mathrm{target}}$ exactly, the initial product model must be calibrated through the linear constraints

$$\sum_i q_i^m B(x_i Y_0, K_n, w(0,T_m)) = C_n^{m,\mathrm{target}}, \qquad \sum_i q_i^m = 1, \qquad \sum_i q_i^m x_i = X_0, \qquad q_i^m \geq 0, \quad (24)$$

together with martingale-transport consistency across m. If these constraints are feasible, the theorem supplies an exact dynamically arbitrage-free smooth extension of that grid. If they are infeasible for the chosen anchors and smoothing clock, one must move or enlarge the anchors, reduce the smoothing, or fit within bid–ask. The drift proof cannot repair a failed static inverse problem.

7 Front expiry and the removal of observed-spot pinning

The construction has a particularly clean front-expiry interpretation. The latent factor X_{T_1} lands on the finite DLV support, but the observed spot is

$$S_{T_1} = X_{T_1} Y_{T_1}.$$

For a nondegenerate Black factor, S_{T_1} has continuous support and is not pinned to the fixed cash strike grid. Nevertheless the call settles exactly, because $w(t,T_1) \downarrow 0$ and the surviving moving anchor is $X_{T_1} Y_{T_1} = S_{T_1}$.

Thus the product construction separates two questions:

1. the DLV factor X carries the finite martingale transport and obeys the Wiese–Wissel drift condition;
2. the Black factor Y turns its atoms into a smooth observable call distribution while preserving one common underlying and ordinary settlement.

It is an exact front-expiry construction, not merely a terminal interpolation. Its empirical adequacy and the feasibility of the inverse calibration (24) are separate numerical questions.

8 Conclusion

The exact theorem can be summarised as

$$\boxed{\begin{array}{c} \text{coherent DLV martingale } X \quad + \quad \text{independent Black martingale } Y \\ \Longrightarrow \quad S = XY, \qquad C_t(T,K) = \displaystyle\sum_i q_i(t,T) B(x_i Y_t, K, V(T) - V(t)), \\ \mathrm{drift}\, C_t(T,K) = 0 \quad \text{for every fixed } (T,K). \end{array}} \quad (25)$$

On the X grid, dynamic no-arbitrage is encoded by $\alpha = -\eta \cdot \beta$. Between grid points, it is encoded by the Black heat equation together with orthogonality of the X and Y martingale parts. The two cancellations are not competing formulas: they are the grid and smooth parts of the same product construction.

References

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